EC991 Topics in Macroeconomics

Week 4 · Computing the Aiyagari Equilibrium

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Algorithm in a nutshell

The solution is characterised by a **fixed point** in the interest rate:

Step 1. Guess an interest rate r_0 (or a quantity of capital K_0)

Step 2. Use r_0 in the firm's first-order conditions to recover capital demand and the wage:

$$K(r_0) \;=\; \left(\frac{\alpha}{r_0+\delta}\right)^{\frac{1}{1-\alpha}} N, \qquad w(r_0) \;=\; (1-\alpha)\left(\frac{K(r_0)}{N}\right)^{\alpha}$$

Step 3. Given w and r_0, solve the household problem for the policy function $g_a(a, y;\, r_0)$:

$$V(a,y) \;=\; \max_{a' \geq -\phi} \left\{ u\big(Ra + wy - a'\big) \;+\; \beta \sum_{y'} P(y' \mid y)\, V(a', y') \right\}$$

Step 4. Given g_a, find the **stationary distribution**, the Λ that replicates itself under the transition function Q:

$$\Lambda(\mathcal{A} \times \mathcal{Y}) \;=\; \int_{A \times Y} Q\big((a,y),\, \mathcal{A} \times \mathcal{Y}\big)\, d\Lambda(a,y)$$

Step 5. Given Λ, compute **excess demand**, asset supply minus capital demand:

$$\Phi(r_0) \;=\; \int_{A \times Y} a \; d\Lambda(a, y;\, r_0) \;-\; K(r_0)$$

If $\Phi(r_0) = 0$: equilibrium found. Otherwise update the guess r_0 and return to step 2

Week 3 tutorial: what we built

Asset grid.

- n_a points from $a_1 = -\phi$ to a_{n_a}, with the upper bound chosen so that it never binds
- More points near the borrowing limit, where the policy bends: growth-rate spacing with rate ν

Income process.

- The AR(1) for $\log y$ becomes a grid $y_1, \ldots, y_{n_y}$ and a transition matrix P
- **Tauchen** matches the conditional distribution, but leaves gaps when ρ is high
- **Rouwenhorst** handles persistent processes

Value function iteration.

- The Bellman operator is a contraction: iterate $V^n \mapsto V^{n+1}$ until $\max_{i,j} |V_{ij}^{n+1} - V_{ij}^n| < \text{tol}$
- **Grid search**: robust, but the policy lives on the grid and the method is slow
- **Interpolate, then optimise**: a' can fall between grid points; more accurate for the same n_a

Interpolation.

- Linear (simple, corners), cubic spline (smooth, can overshoot), Chebyshev (accurate when smooth, poor at kinks)
- Practical check: add grid points and confirm the saving policy does not move

Today

1.	**Faster household solutions**	Howard's improvement, monotonicity, the endogenous grid method
2.	**How accurate is the solution?**	Euler equation errors
3.	**The stationary distribution**	a histogram on the grid, moved by a lottery between grid points
4.	**The equilibrium interest rate**	aggregate asset supply, excess demand, root finding
5.	**What the model delivers**	the model's wealth distribution against the data
6.	**Policy**	progressive taxation, and how to evaluate it

Speeding up VFI: Howard's improvement algorithm

- **Where the time goes.** Every VFI iteration solves a maximisation at each grid state (a_i, y_j): the search over a' is the expensive step
- **What we observe.** The saving policy g^n often settles down long before the value function V^n does, so later sweeps repeat the same search and return the same answer
- **The question.** Once the policy has settled, can we stop paying for the search and only update the value?
- **Howard's idea**: keep the saving rule fixed for several cheap value updates, then maximise again using the improved value estimate

Improve the policy $\longrightarrow$ Evaluate it n_H times $\longrightarrow$ Improve the policy again

- **Improve**: iteration n is one maximisation sweep over every grid state (a_i, y_j), giving policy g^n_{ij} and value V^n_{ij}
- **Evaluate**: set $W^0 = V^n$, hold g^n fixed, and update the value at every state, with no maximisation:

$$W^{k+1}_{ij} \;=\; u\big(Ra_i + wy_j - g^n_{ij}\big) \;+\; \beta \sum_m P_{jm}\, W^k\big(g^n_{ij},\, y_m\big), \qquad k = 0, \ldots, n_H - 1$$

  W^k is the value estimate after k updates; if g^n_{ij} falls between grid points, interpolate W^k

- **Repeat**: maximise again using W^{n_H}, and check convergence
- **We choose** n_H, trading cheap evaluation updates against expensive maximisation sweeps; $n_H = 0$ is ordinary VFI

Speeding up VFI: exploiting monotonicity and concavity

Howard reduces **how often** we maximise. These shortcuts make each maximisation **cheaper**: fewer saving choices to try

$$\max_{a' \text{ feasible}} F(a';\, a_i, y), \qquad F(a';\, a_i, y) \;=\; u(Ra_i + wy - a') + \beta \sum_{y'} P(y' \mid y)\, V(a', y')$$

Monotone policy: start at the previous solution. If $g_a(a_i, y) \geq g_a(a_{i-1}, y)$ at each income y, work up the asset grid. For example, on the grid $a' \in \{0, 1, 2, 3, 4, 5\}$, if the best choice at a_{i-1} was 3, then at a_i we search only $\{3, 4, 5\}$

Concave objective: stop at the first fall. If F is concave in a', its slope only falls, so once F drops it never rises again: the first fall marks the maximum. In the table, after $a' = 4$ we stop and pick $a' = 3$, so $a' = 5$ is never evaluated

Saving choice a'	0	1	2	3	4	5
Objective F	10	14	17	**18**	17.5	–

Together. Start at the previous state's best choice, then stop at the first fall

With interpolation. The previous solution is a lower bound for the optimiser, and under concavity any local maximum is global. The interpolated objective must keep these properties: artificial wiggles can break concavity

At scale. The divide-and-conquer algorithm of Gordon and Qiu (2018, QE) exploits monotonicity further

Policy function methods

- VFI is **robust**: it works under a wide set of conditions (discrete choices, several controls, …)
- But it tends to be **slow**, and often not very accurate
- **Policy function methods** iterate on the Euler equation instead: fast and accurate, but less robust
- We study the **endogenous grid method** (EGM), probably the most used today

Time iteration on the Euler equation

- Let $c = g_c(a, y)$ be the consumption policy. With CRRA utility, $u'(c) = c^{-\gamma}$, and the Euler equation is a **functional equation** in g_c:

$$c^{-\gamma} \;=\; \beta R\, \mathbb{E}\big[g_c(a', y')^{-\gamma} \,\big|\, y\big] \;=\; \beta R\, \mathbb{E}\Big[g_c\big(Ra + wy - c,\; y'\big)^{-\gamma} \,\Big|\, y\Big]$$

  the second form substitutes the budget constraint $a' = Ra + wy - c$

- **Time iteration** (Coleman, 1990): treat the guess for g_c as tomorrow's policy, and at every grid state today solve for the c that satisfies the equation, with interpolation and a **root finder**
  - the solutions form the new table g_c^{n+1}; if it is not close enough to g_c^n, it becomes tomorrow's guess and we repeat
- A non-linear equation solver at every state, every iteration, is **costly**
- Carroll (2005): the **endogenous grid method** bypasses the root finder
  - Barillas and Fernández-Villaverde (2007) extend it to combine VFI and EGM

Endogenous grid method: the idea

- a' is the assets the household **carries into tomorrow**
- **Root finding** fixes today's assets a and searches for c. Each trial c moves $a' = Ra + wy - c$, so tomorrow's consumption must be recomputed
- **EGM** works backwards: fix tomorrow's assets a', calculate today's consumption directly, then recover the assets $\hat{a}$ the household must have held today
- With a' fixed, the right-hand side of the Euler equation does not depend on the unknown c: no root finder
- We reuse the same asset-grid numbers, but **change their role**: they are now tomorrow's assets, and the method generates today's

Endogenous grid method: an example (1/2)

Start. Asset grid $\{0, 1, 2, 3, 4\}$, income grid $\{y_1, \ldots, y_5\}$. The guess $g_c^n(a_i, y_j)$ is a table of 25 consumption values

Pick $a' = 2$ and $y = y_3$. We have **not** fixed today's assets

Look up tomorrow's consumption at $a' = 2$ in all five income states, weighted by their probabilities given y_3. Everything is known:

$$\text{RHS}(2, y_3) \;=\; \beta R \sum_{m=1}^{5} P_{3m} \left[g_c^n(2, y_m) \right]^{-\gamma}$$

y_3 enters only through the weights P_{3m}. Tomorrow's consumption is evaluated at tomorrow's income y_m, inside the sum

Invert the Euler equation $c^{-\gamma} = \text{RHS}(2, y_3)$ to get consumption today directly:

$$\tilde{c} \;=\; \text{RHS}(2, y_3)^{-1/\gamma}$$

Recover today's assets from the budget constraint $\tilde{c} + 2 = R\hat{a} + wy_3$:

$$\hat{a} \;=\; \frac{\tilde{c} + 2 - wy_3}{R}$$

With numbers. Suppose $\tilde{c} = 1.5$, $R = 1.02$, $wy_3 = 1$: then $\hat{a} = 2.5/1.02 \approx 2.451$, so $g_c^{n+1}(2.451, y_3) = 1.5$. Consuming 1.5 there leaves exactly 2

Endogenous grid method: an example (2/2)

Repeat for all 25 pairs (a_i', y_j). Each one gives a consumption value and a current-asset value:

$$\tilde{c}_{ij} \quad \text{and} \quad \hat{a}_{ij} \;=\; \frac{\tilde{c}_{ij} + a_i' - w y_j}{R}$$

The two subscripts on $\hat{a}_{ij}$ matter: the generated asset values generally differ across income states

Interpolate back to the original grid. At y_3 we have five pairs, $(\hat{a}_{13}, \tilde{c}_{13}), \ldots, (\hat{a}_{53}, \tilde{c}_{53})$. They describe the updated policy, but at the generated asset values, not generally at $0, 1, 2, 3, 4$

Interpolate through them and evaluate at $a = 0, 1, 2, 3, 4$: this gives $g_c^{n+1}(0, y_3), \ldots, g_c^{n+1}(4, y_3)$

Do this separately for all five income states: the updated 5×5 table on the **original grid**

Compare old and new entries, now at the same states:

$$d \;=\; \max_{i,j} \left| g_c^{n+1}(a_i, y_j) - g_c^{n}(a_i, y_j) \right|$$

Interpolation does not cover grid points below the smallest $\hat{a}_{ij}$ at that income: see the borrowing constraint slide

Endogenous grid method: the algorithm (1/2)

Step 1. Guess a consumption policy on the **exogenous** grid, the one we discretised at the start: $c = g_c^n(a_i, y_j)$

Step 2. Use the guess to compute the right-hand side of the Euler equation. We now treat the grid point as **tomorrow's** assets, a_i':

$$\text{RHS}(a_i', y_j) \;=\; \beta R \sum_{m=1}^{n_y} P_{jm}\; u'\big(g_c^n(a_i', y_m)\big)$$

Step 3. Invert marginal utility to get the consumption **today** that goes with saving a_i':

$$\tilde{c} \;=\; u'^{-1}\big(\text{RHS}(a_i', y_j)\big)$$

- $\tilde{c}$ is the next iteration's policy, $\tilde{c} = g_c^{n+1}(\hat{a}_i, y_j)$, at the state $(\hat{a}_i, y_j)$ from which the household saves $a_i' = g_a^n(\hat{a}_i, y_j)$
- But we **cannot** compare $g_c^{n+1}(\hat{a}_i, y_j)$ with $g_c^n(a_i, y_j)$: they live on **different asset grids**, $\hat{a}_i \neq a_i$

Endogenous grid method: the algorithm (2/2)

Step 4. Put g_c^{n+1} back on the grid of g_c^n. First find the **endogenous grid** $\hat{a}_i$ from the budget constraint:

$$\hat{a}_i \;=\; \frac{\tilde{c}(a'_i, y_j) + a'_i - w\, y_j}{R}$$

Step 5. Interpolate through the pairs $\left(\hat{a}_i,\ g_c^{n+1}(\hat{a}_i, y_j)\right)$ to get $g_c^{n+1}(a_i, y_j)$ on the exogenous grid

Step 6. Compute the distance $d = \max_{i,j} \left| g_c^{n+1}(a_i, y_j) - g_c^n(a_i, y_j) \right|$. If $d <$ tol: done.
Otherwise update the guess and start again

- No maximisation and no root finding: every step is an evaluation, an inversion or an interpolation

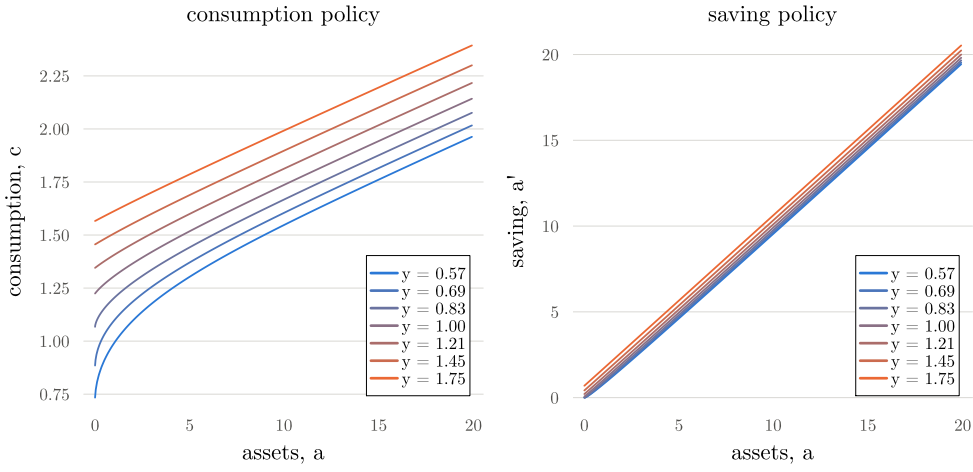
Endogenous grid method: the borrowing constraint

- To handle the borrowing constraint, it is often easier to interpolate the **asset policy** instead: $a_i' = g_a(\hat{a}_i, y_j)$
- After interpolating to get $g_a(a_i, y_j)$ on the exogenous grid, check whether the constraint binds, $g_a(a_i, y_j) < a_1$:
  - if $g_a(a_i, y_j) < a_1$: set $g_a(a_i, y_j) = a_1$
  - if $g_a(a_i, y_j) \geq a_1$: do nothing
- After this correction, recover consumption from the budget constraint and proceed as usual:

$$g_c^{n+1}(a_i, y_j) \;=\; R\, a_i + w\, y_j - g_a(a_i, y_j)$$

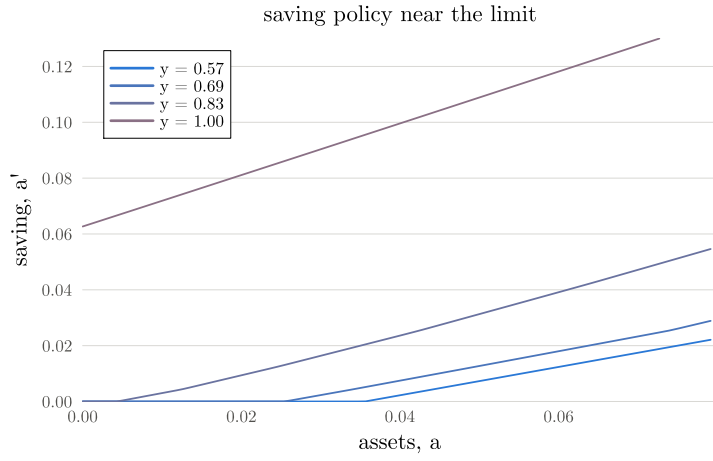
recall $a_1 = -\phi$: the bottom of the grid is the borrowing limit

Endogenous grid method: policy functions



- One line per endowment: consumption (left) and savings (right) rise with assets at every income level

Endogenous grid method: policy functions near the limit



- Near the limit, low-income households save nothing until their assets pass a threshold; higher-income households, whose lines start above this window, always save

Endogenous grid method: why is it less robust?

Basic EGM gains its speed from the structure of the standard Aiyagari household problem. Our calculation ran:

$$\text{pick } a' \longrightarrow \text{calculate } c \text{ from the Euler equation} \longrightarrow \text{recover current assets } \hat{a} \longrightarrow \text{interpolate}$$

Three things can complicate this sequence:

1. **The Euler equation may not identify the best decision.** With a fixed fee for changing assets, the household asks whether to pay and how much to save. The Euler equation helps only with the second question; for the first we compare the values of adjusting and not adjusting

2. **Calculating consumption may become harder.** In our example $c^{-\gamma} = \text{RHS}$ gives $c = \text{RHS}^{-1/\gamma}$ at once. With more complicated preferences, solving $u'(c) = \text{RHS}$ may need a numerical solver, and part of the speed gain disappears

3. **Constraints need rules beyond the Euler equality.** At the borrowing limit we impose $a' = a_1$ and recover consumption from the budget constraint. One constraint is straightforward; with several, finding which ones bind is harder

Our model has this structure. More complicated models can need extensions, whereas VFI compares the values of feasible choices more directly

Euler equation errors

- To compare the **accuracy** of different solution methods, compute the **Euler equation errors**
- Define the approximation error ε as the share of today's consumption that would close the Euler equation:

$$u'\big(c_t(1-\varepsilon)\big) \;=\; \beta R\,\mathbb{E}_t\big[u'(c_{t+1})\big] \qquad \Longleftrightarrow \qquad \varepsilon \;=\; 1 - \frac{u'^{-1}\big(\beta R\,\mathbb{E}_t[u'(c_{t+1})]\big)}{c_t}$$

- Compute ε on a **different grid** from the one used to solve the model, or along a long simulated history of shocks
- Only where the constraint is slack: at the borrowing limit the Euler equation holds as an inequality
- Application: Aruoba, Fernández-Villaverde and Rubio-Ramírez (2006, JEDC)

Euler equation errors: an example

How well does our solved 5×5 consumption table satisfy the Euler equation? Illustrative numbers: log utility, $\beta = 0.95$, $R = 1$

Pick one state: $a = 2$, $y = y_3$, with $wy_3 = 1$. The table says $g_c(2, y_3) = 2$, so the household saves $a' = 2 + 1 - 2 = 1$. With a zero borrowing limit, the choice is unconstrained

Look up tomorrow's consumption at $a' = 1$ in all five income states, with probabilities from row 3 of P:

Tomorrow's income	y_1	y_2	y_3	y_4	y_5
Consumption $g_c(1, y_m)$	1.5	1.8	2.0	2.2	2.5
Probability P_{3m}	0.2	0.2	0.2	0.2	0.2

Calculate the consumption the Euler equation implies, c^{EE}, from tomorrow's expected marginal utility:

$$\frac{1}{c^{EE}} \;=\; 0.95 \times 0.2 \left(\frac{1}{1.5} + \frac{1}{1.8} + \frac{1}{2} + \frac{1}{2.2} + \frac{1}{2.5} \right) \;\approx\; 0.4896 \qquad \Longrightarrow \qquad c^{EE} \;\approx\; 2.0425$$

Compare it with what the table prescribes: $\varepsilon = 1 - 2.0425/2 \approx -0.0213$, an error of about 2.13% of consumption

Repeat at the other unconstrained states and report, say, the largest absolute error. To test interpolation too, check states between grid points, such as $(2.4, y_3)$

We compare today's computed consumption with c^{EE}, not with consumption tomorrow, which can legitimately differ

Invariant distribution

- With the household's policy functions, we compute the **invariant distribution** $\lambda(a, y)$: step 4
- Three methods:
  1. Monte Carlo simulation
  2. **Non-stochastic simulation**
  3. Parameterised distributions
- We use **method 2**
  - method 1 is usually too slow
  - method 3 suits aggregate uncertainty, but is cumbersome (Algan, Allais and den Haan, 2008)

Invariant distribution: a histogram on a grid

- Approximate the distribution by a **histogram** over a fixed grid (Young, 2010, JEDC)
- The asset grid must be **fine**. It need **not** be the grid used for the value or policy function
- The distribution is stored in an array with $n_a \times n_y$ entries, $\lambda(a_i, y_j)$: a matrix $n_a \times n_y$, or a vector of length $n_a n_y$
- Then build the **transition function** Q of week 3: the probability of moving from state (a, y) to state (a', y'):

$$\begin{aligned} Q((a,y),\,(a',y')) &= \Pr\left[a_{t+1} = a',\ y_{t+1} = y' \,\middle|\, a_t = a,\ y_t = y\right] \\ &= \Pr\left[a_{t+1} = a' \,\middle|\, a_t = a,\ y_t = y\right] \times P(y' \mid y) \end{aligned}$$

they multiply: given today's state, the savings choice and tomorrow's endowment are independent

Invariant distribution: the law of motion on the grid

- The transition function and the histogram define a **Markov chain**: in week 3's matrix form, $\lambda_{t+1} = Q'\lambda_t$
- From a distribution $\lambda_t(a, y)$, the mass at node (a_k, y_m) next period is

$$\lambda_{t+1}(a_k, y_m) \;=\; \sum_{i=1}^{n_a} \sum_{j=1}^{n_y} \lambda_t(a_i, y_j)\; Q\big((a_i, y_j),\, (a_k, y_m)\big)$$

- Computing Q needs the savings policy $g_a(a, y)$ and the transition matrix P:

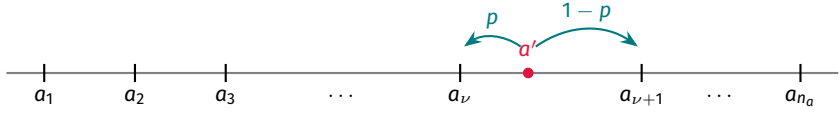
$$Q\big((a_i, y_j),\, (a_k, y_m)\big) \;=\; \underbrace{\Pr\left[a_{t+1} = a_k \,\middle|\, a_t = a_i,\; y_t = y_j\right]}_{\text{savings decision}} \times \underbrace{P_{jm}}_{\text{exogenous income shocks}}$$

Invariant distribution: savings between grid points

- The policy gives $a' = g_a(a_i, y_j)$. Problem: a' usually lies **off the grid**
  - in week 3 the policy was on the grid, so the savings probability was an indicator, $\mathcal{I}\{g_a(a, y) = a'\}$
- The trick: send some households to the grid point below and some above, so that the **mass of assets is preserved**:

$$a' \;=\; p\, a_\nu + (1-p)\, a_{\nu+1}, \qquad p \;=\; \frac{a_{\nu+1} - a'}{a_{\nu+1} - a_\nu}$$

where a_ν is the grid point just below a'



- $\Pr[a_{t+1} = a_\nu] = p$, $\Pr[a_{t+1} = a_{\nu+1}] = 1 - p$, and $\Pr[a_{t+1} = a_k] = 0$ for every other k

Invariant distribution: solving for it

- Multiply the lottery probabilities $\Pr[a_{t+1} = a_k \mid a_i, y_j]$ by P_{jm} and we have the transition function Q
- This is just a Markov chain, so the stationary distribution solves a **linear system**, $\lambda = Q'\lambda$, by standard methods
  - each row of Q has at most $2n_y$ non-zero entries: Q is a **sparse** matrix
- **Iteration** tends to be robust:
  - **Step 1.** Guess $\lambda^n(a, y)$: a uniform mass, or anything that sums to one
  - **Step 2.** Apply the transition function to get next period's distribution, $\lambda^{n+1}(a, y)$
  - **Step 3.** Compute $d = \max_{i,j} \left|\lambda^{n+1}(a_i, y_j) - \lambda^n(a_i, y_j)\right|$. If $d <$ tol: done. Otherwise use λ^{n+1} as the new guess and repeat

Invariant distribution: an example

Asset grid $\{0, 1, 2, 3, 4\}$, income grid $\{y_1, \ldots, y_5\}$: 25 states, so λ is a 5×5 table and Q is 25×25. Unlike the policy table, each entry of λ is a **share of the population**: $\lambda(2, y_3) = 0.10$ means 10% of households hold assets 2 in income state y_3

Where do they go? Suppose the policy at $(2, y_3)$ gives $a' = 1.3$, between the nodes 1 and 2. The lottery splits the mass: a share $p = (2 - 1.3)/(2 - 1) = 0.7$ goes to node 1, and 0.3 to node 2, since $0.7 \times 1 + 0.3 \times 2 = 1.3$

Build Q. Multiply by the income probabilities. With $P_{3m} = 0.2$ for every m:

$$Q\big((2, y_3), (1, y_m)\big) \;=\; 0.7 \times 0.2 \;=\; 0.14, \qquad Q\big((2, y_3), (2, y_m)\big) \;=\; 0.3 \times 0.2 \;=\; 0.06$$

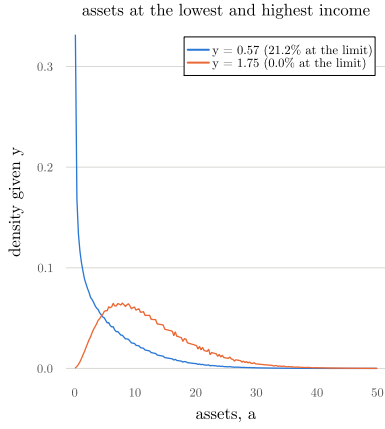
and every other entry of that row of Q is zero: at most $2n_y = 10$ destinations out of 25

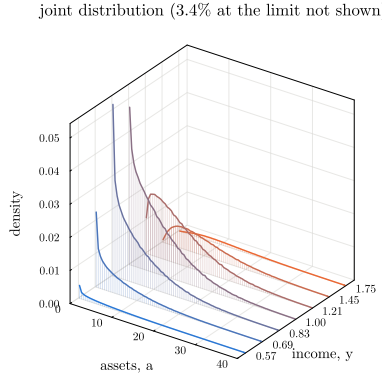
Move the mass. The 10% at $(2, y_3)$ send $0.10 \times 0.14 = 0.014$ to each $(1, y_m)$ and $0.10 \times 0.06 = 0.006$ to each $(2, y_m)$. Add up such contributions from all 25 starting states to get λ_{t+1}: that is $\lambda_{t+1} = Q'\lambda_t$

Iterate. Start from $\lambda^0(a_i, y_j) = 1/25$ and apply Q' until the table stops changing. Households keep moving, but the share at each state no longer does: $\lambda = Q'\lambda$. The policy, prices and Q stay fixed throughout; we iterate on the distribution, not on the household problem

The lottery is an approximation: it preserves total mass and mean assets at each transition, not the whole distribution

Invariant distribution





- Left: assets at the lowest and highest endowment; low-income households pile up at the borrowing limit
- Right: the joint distribution, $\lambda(a, y)$ per unit of assets, one curve per income state: most households have middle incomes, and wealth rises with income

Aggregate asset supply

- With the invariant distribution $\lambda(a, y)$, we compute **aggregate asset supply**
  - on the grid, the integral $\int a \, d\Lambda$ becomes a sum, and the weight $d\Lambda$ becomes the histogram λ:

$$\mathbb{E}a = \sum_{i=1}^{n_a} a_i \sum_{j=1}^{n_y} \lambda(a_i, y_j)$$

- Together with capital demand $K(r)$, this gives the **equilibrium condition**
- We can now write an iterative procedure that finds the **steady-state equilibrium**

Finding the equilibrium: setup and prices

Set up once, before the interest rate loop

- **Income**: Tauchen or Rouwenhorst gives the states y_j and the transition matrix P
  - π, the stationary distribution of the endowment chain, gives labour supply $N = \sum_j y_j \pi_j$
- **Assets**: choose the borrowing limit, an upper bound that does not bind, and the grid spacing
  - a **policy grid** A^{pol}, with more points near the borrowing limit
  - a **distribution grid** A^{dist}, possibly finer; both use the same income states

Step 1. Guess an interest rate r^n

- As we saw earlier, the equilibrium r lies between $-\delta$ and $1/\beta - 1$: these bounds are the initial bracket

Step 2. Use the firm's first-order conditions to compute capital demand and the wage:

$$K(r^n) \;=\; \left(\frac{\alpha}{r^n + \delta}\right)^{\frac{1}{1-\alpha}} N, \qquad w(r^n) \;=\; (1-\alpha)\left(\frac{K(r^n)}{N}\right)^{\alpha}$$

- The income process, the grids and N stay **fixed** as we update r

Finding the equilibrium: the household problem

Step 3. Given r^n and $w(r^n)$, compute the saving policy $g_a(a, y;\ r^n)$ on $A^{\text{pol}} \times Y$, by either method:

- **VFI**: maximise the Bellman objective at every state, by grid search or by interpolating V and optimising
  - Howard: choose n_H, the fixed-policy updates between maximisation sweeps
  - where valid, policy monotonicity and objective concavity shorten the search
- **EGM**: fix tomorrow's assets, invert the Euler equation, recover today's assets
  - the generated asset values differ across income states and iterations
  - interpolate the saving policy back onto the grid, enforce $a' \geq -\phi$, recover consumption from the budget constraint
- Both methods deliver the **same object** that step 4 needs: the saving policy

Check the household solution

- Iterate until the value or the policy converges
- Euler equation errors away from the borrowing constraint, including between grid points
- Refining the grid should not materially change the policy

Finding the equilibrium: distribution and market clearing

Step 4. Compute the stationary distribution on $A^{\text{dist}} \times Y$, with prices and the saving policy fixed:

- Interpolate the saving policy onto A^{dist}; split mass between the two nodes around a'
- Multiply those weights by P_{jm} to get Q, then iterate $\lambda^{\ell+1} = Q'\lambda^{\ell}$ until λ converges

Step 5. Aggregate assets and compute **excess demand**, asset supply minus capital demand:

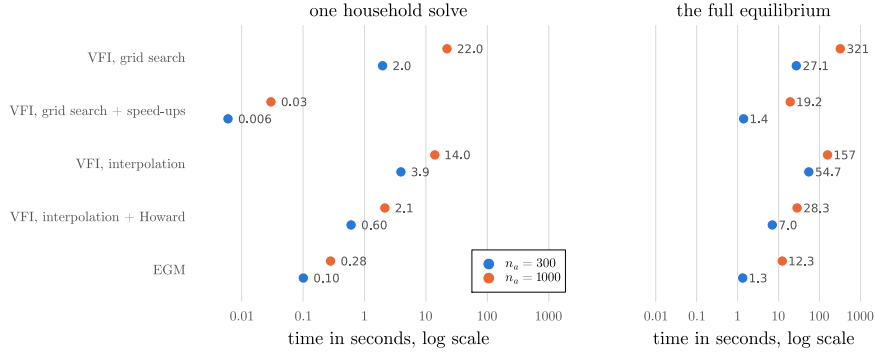
$$\Phi(r^n) = \underbrace{\sum_{i,j} a_i \, \lambda(a_i, y_j;\, r^n)}_{\text{asset supply } \mathbb{E}a(r^n)} - \underbrace{K(r^n)}_{\text{capital demand}}, \qquad a_i \in A^{\text{dist}}$$

Step 6. If $|\Phi(r^n)| <$ tol: we have found an equilibrium. Otherwise update r by bisection or Brent's method:

- if $\Phi(r^n) > 0$: households supply more assets than firms demand. **Decrease** r
- if $\Phi(r^n) < 0$: firms demand more capital than households supply. **Increase** r
- Return to step 2: recompute prices, the saving policy and the distribution

More details

Speed across methods



- The axis is a log scale: each gridline is ten times slower than the one before
- **Speed-ups matter most for plain grid search**: at $n_a = 1000$ the equilibrium falls from 321 to 19 seconds, with the same answer
- **One household solve flatters grid search**: it is quickest there, but the equilibrium also needs the distribution, which converges slowly on a policy on the grid
- **Speed is not the whole story**: grid search's r^* differs from the others'. EGM is the quickest method that gets it right

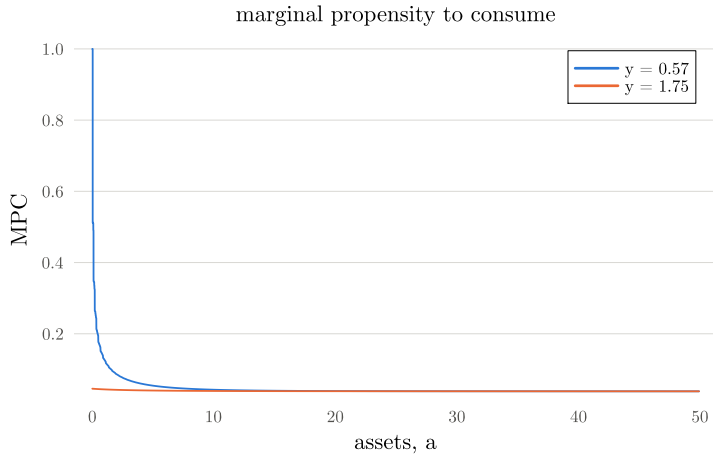
One thread on a 2.2 GHz server, after a warm-up. Your times will differ

Model statistics

Prices and aggregates		Distribution of assets	
Interest rate r	0.0377	Mean	7.422
Wage w	1.2870	Standard deviation	6.729
Capital K	7.423	25th percentile	2.153
Asset supply $\mathbb{E}a$	7.422	Median	5.686
Labour supply N	1.027	95th percentile	20.745
K/N	7.230	99th percentile	28.923
Output Y	1.972	Gini	0.483
Assets to output	3.764	Top 10% share	0.299
Constrained households	0.034	Top 1% share	0.045

- Equilibrium prices, aggregates, the share of constrained households (those choosing $a' = -\phi$) and the wealth distribution
- $\beta = 0.96$, $\gamma = 2$, $\phi = 0$, $\alpha = 0.33$, $\delta = 0.05$, $\rho = 0.9$, $\sigma = 0.1$; solved by EGM with $n_a = 1000$ and $n_y = 7$

Model statistics: marginal propensities to consume



- The MPC is high near the borrowing limit and falls quickly with wealth

Steady-state determination

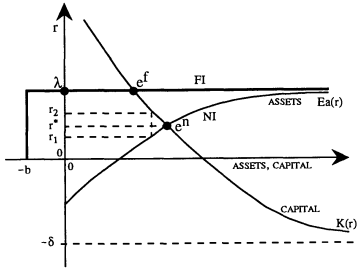
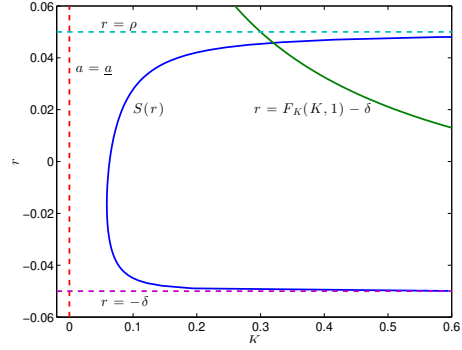


FIGURE IIb
Steady-State Determination

(A) Aiyagari (1994)



(B) A more precise version

- To trace $\mathbb{E}a(r)$: at each r, feed r and $w(r)$ into the household problem, find λ, and add up assets
- The equilibrium is where $\mathbb{E}a(r)$ crosses capital demand $K(r)$; Aiyagari's λ in (A) and Moll's ρ in (B) are $1/\beta - 1$, and $S(r)$ in (B) is our $\mathbb{E}a(r)$

Aiyagari 1994, Figure IIb, and Moll, EC442 Lecture 3, slides 30 and 31

Comparing model and data

The model's wealth distribution

Discount factor	0.945	0.95	**0.955**	0.96	0.97
Var log gross labor inc	0.982	0.982	**0.982**	0.982	0.982
Gini gross labor inc	0.505	0.505	**0.505**	0.505	0.505
Var log net labor inc	0.982	0.982	**0.982**	0.982	0.982
Gini net labor inc	0.505	0.505	**0.505**	0.505	0.505
Var log consumption	0.987	0.980	**0.966**	0.941	0.833
Gini consumption	0.497	0.493	**0.486**	0.476	0.443
Mean wealth	1.414	2.067	**3.053**	4.599	12.003
Median wealth	0.017	0.130	**0.379**	0.930	4.935
Gini wealth	0.858	0.831	**0.799**	0.762	0.662
P90-P50 wealth	220	44	**23**	14	7
P99-P50 wealth	1217	209	**94**	50	17
Frac wealth<=0	47%	30%	**25%**	20%	6%
Frac wealth<= 5% $E[y]$	53%	46%	**39%**	26%	11%
Top 10% wealth share	75%	70%	**64%**	59%	47%
Top 1% wealth share	22%	19%	**17%**	14%	9%

- Less patient households put less weight on future consumption, so they save less and run down their buffers more readily
- More households end up near zero wealth
- Wealth becomes more concentrated among those who still hold assets, even if those households also become poorer
- Example: two households hold £10 and £90, so the richer owns 90%. If their wealth falls to £1 and £49, the richer owns 98%, despite having less wealth

Wealth in the data: SCF 2016

Full Distribution	Total	Non-Housing	Financial
Mean	11.19	8.48	5.34
Median	1.58	0.49	0.31
P90	19.3	12.6	9.6
P99	168.0	141.4	87.0
P99.9	700.5	634.6	358.9
P90-P50 Ratio	12	26	31
P99-P50 Ratio	106	288	284
Top 10% Share	77%	84%	81%
Top 1% Share	39%	45%	40%
Top 0.1% Share	15%	18%	14%
Gini	0.86	0.91	0.89
Frac <=0	11%	15%	11%
Frac <= $2000	17%	22%	31%
Frac <= 5% Av Earns	18%	23%	34%

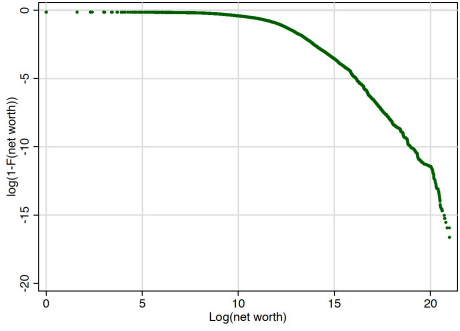
Survey of Consumer Finances 2016; Moll, EC442 Lecture 3

Wealth in the data: SCF 2016 without the top 1%

Excluding top 1%	Total	Non-Housing	Financial
Mean	7.79	5.32	3.95
Median	1.71	0.53	0.33
P90	19.8	12.6	9.7
P99	106.0	85.1	56.9
P99.9	179.1	163.1	133.5
P90-P50 Ratio	12	24	30
P99-P50 Ratio	62	160	174
Top 10% Share	65%	74%	73%
Top 1% Share	18%	23%	23%
Top 0.1% Share	2%	3%	4%
Gini	0.79	0.86	0.84
Frac <=0	12%	15%	11%
Frac <= $2000	17%	22%	31%
Frac <= 5% Av Earns	18%	23%	33%

Survey of Consumer Finances 2016; Moll, EC442 Lecture 3

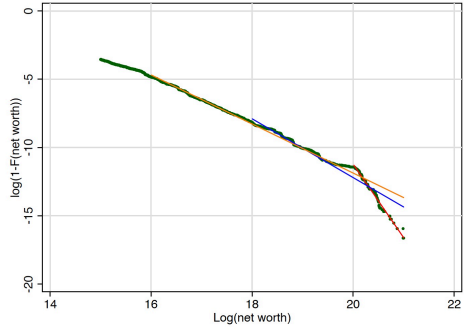
The Pareto tail of wealth: SCF 2013



Survey of Consumer Finances 2013; Moll, EC442 Lecture 3

- Each point: the fraction of households with wealth above a, $1 - F(a)$. A fraction of households, not a share of wealth
- A club with minimum wealth a: raise the bar and fewer households can enter
- The log of a fraction is negative: -2.3 is 10%, -4.6 is 1%, -6.9 is 0.1%
- Flat at low thresholds, where raising the bar excludes few households. Falling at high ones

The Pareto tail of wealth: SCF 2013, the top



Survey of Consumer Finances 2013; Moll, EC442 Lecture 3

- The top of the distribution: close to a straight line up to log wealth of about 20
- Straight line on log axes: multiplying a by a given factor divides the fraction above it by a roughly constant factor
- That is a Pareto tail:

$$\Pr(\text{wealth} > a) = Ca^{-\zeta}$$

$$\log \Pr(\text{wealth} > a) = \log C - \zeta \log a$$

The slope is $-\zeta$

The model's wealth distribution with transitory shocks

Transitory shock size	0.05	0.1	**0.2**	0.25	**0.2**
Discount factor	0.955	0.955	**0.955**	0.955	**0.9**
Var log gross labor inc	0.985	0.992	**1.021**	1.043	**1.021**
Gini gross labor inc	0.509	0.513	**0.521**	0.526	**0.521**
Var log net labor inc	0.985	0.992	**1.021**	1.043	**1.021**
Gini net labor inc	0.509	0.513	**0.521**	0.526	**0.521**
Var log consumption	0.964	0.963	**0.959**	0.957	**1.001**
Gini consumption	0.486	0.486	**0.486**	0.486	**0.513**
Mean wealth	3.071	3.105	**3.212**	3.285	**0.178**
Median wealth	0.399	0.430	0.543	0.617	**0.026**
Gini wealth	0.795	0.790	**0.776**	0.767	**0.814**
P90-P50 wealth	22	21	**17**	15	**16**
P99-P50 wealth	90	83	**66**	59	**90**
Frac wealth<=0	6%	5%	**4%**	3%	**24%**
Frac wealth<= 5% $E[y]$	34%	29%	**20%**	16%	**59%**
Top 10% wealth share	64%	64%	**62%**	61%	**69%**
Top 1% wealth share	16%	16%	**16%**	16%	**25%**
Top 0.1% wealth share	3%	3%	**3%**	3%	**6%**

- **Larger transitory shocks barely move wealth concentration**: a shock that soon fades needs only a small buffer
- **More impatience raises concentration by emptying the bottom**: at $\beta = 0.9$ the top 1% share is 25%, but 24% of households hold no wealth
- **The top stays out of reach**: in the 2016 SCF the top 1% holds 39% and the top 0.1% holds 15%, so the top needs another mechanism

Policy in the Aiyagari model

Extension: progressive taxation

- A functional form that captures progressivity (Benabou, 2002; Heathcote, Storesletten and Violante, 2017):

$$T(y) \;=\; y - \tau_1 y^{1-\tau_2}$$

  where y is individual labour income
  - τ_2 gives the degree of **progressivity**: the elasticity of post-tax to pre-tax income
  - given τ_2, τ_1 shifts the tax function and sets the **average level** of taxation

- Aggregate tax revenue is the integral over all individuals: $T = \int T(y_i)\, di$
  - the government budget balances by shifting g_y (Heathcote et al., 2017) or by an extra lump-sum transfer (Boar and Midrigan, 2022)

Extension: progressive taxation $T(y) = y - \tau_1 y^{1-\tau_2}$

- The tax is **progressive** if the marginal rate exceeds the average rate, $T'(y) > T(y)/y$, at every income
  - the next pound is taxed more heavily than the pounds before it, so the average rate rises with income
  - higher earners keep a smaller share of what they earn: **relative income differences shrink**
- After-tax income is $\tau_1 y^{1-\tau_2}$, so τ_2 sets how strongly differences are compressed:
  - $\tau_2 = 1$: everyone keeps τ_1, whatever they earn $\Rightarrow$ full redistribution
  - $0 < \tau_2 < 1$: progressivity; with $\tau_2 = 0.5$, four times the earnings leaves $4^{0.5} = 2$ times the after-tax income
  - $\tau_2 = 0$: proportional tax, $T'(y) = T(y)/y = 1 - \tau_1$; everyone keeps the same share
  - $\tau_2 < 0$: regressivity; differences widen
- **Break-even income**: $y_{be} = \tau_1^{1/\tau_2}$, where $T(y_{be}) = 0$
  - if $y_i > y_{be}$: household i is a **taxpayer**
  - if $y_i < y_{be}$: household i **receives a transfer**

How to evaluate optimal policy?

- Suppose we want to compare two tax levels, τ_0 and τ_1:
  - **representative agent**: compare the RA's utility under each
  - **heterogeneous agents**: there is a **distribution** of welfare; we must specify a **social welfare function**
- The most common is **utilitarian**; see Boar and Midrigan (2022) and Bénabou (2002) for discussion
- Compute average lifetime utility, weighted by the distribution, under each policy:

$$W(\tau) = \int_{A\times Y} V(a, y; \tau)\, d\Lambda, \qquad V(a, y; \tau) = \mathbb{E}\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma}}{1-\gamma} \text{ s.t. the budget constraint}$$

How to evaluate optimal policy?

- Comparing policies must account for risk, the endogenous distribution, curvature of utility, ... $\Rightarrow$ use the **consumption-equivalent variation (CEV)**
- CEV: the percentage change Δ in **every** household's consumption that makes it indifferent between the two policies, $W(\tau_0) = W(\tau_1, \Delta)$, where

$$W(\tau_1, \Delta) \;=\; \int \mathbb{E} \sum_t \beta^t \, \frac{\left(c_t^*(1+\Delta)\right)^{1-\gamma}}{1-\gamma} \, d\Lambda \;=\; (1+\Delta)^{1-\gamma} \, W(\tau_1)$$

- If $\Delta > 0$, average welfare is **higher** under policy τ_0:

$$W(\tau_0) = W(\tau_1, \Delta) \qquad \Longleftrightarrow \qquad \Delta \;=\; \left(\frac{W(\tau_0)}{W(\tau_1)}\right)^{1/(1-\gamma)} \;-\; 1$$

Finding the equilibrium: bounds and an alternative

- Finding an equilibrium comes down to finding a **root** of the excess demand function
  - bracketing methods usually work well: bisection, Brent's method, …
- **Initial bracket**: theory gives good bounds for r
  - upper bound: $r_u = 1/\beta - 1$ (with income risk, households would save without limit at this rate)
  - lower bound: $r_l = -\delta$ (capital demand is unbounded as $r \to -\delta$)
- **Alternative**: iterate on K, which gives r and w through the firm's first-order conditions
  - update the capital guess K^n by

$$K^{n+1} \;=\; \omega\, \mathbb{E}a + (1-\omega)\, K^n$$

  - where $\omega \in (0, 1]$ is a **dampening** parameter

◂ Back